

Second-Price Sealed-Bid Auction Simulation Report

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Execution Summary

- Mechanism: Highest bidder wins, pays second highest bid.
- Bidding rule: truthful bidding (Bid = Valuation).
- Parameters: seed=42, auctions=100, bidders=4
- Quality: Uniform[0.0, 100.0]
- Scenarios (sigma): 1:10, 2:30
- Reserve price: 0.0
- Total output rows: 800

Antigravity Demo

- Requested: True
- Success: True
- intercepted open: <https://xkcd.com/353/>

OLS Regression (Price on Quality) with HC1 robust standard errors

_cons	coef=1.641023	robust_se=1.480736	t=1.1082	p=0.267755
Quality	coef=0.251558	robust_se=0.036666	t=6.8608	p=0.000000

Winner's Curse (Overbid) by Scenario

scenario 1: n=400, mean=2.7805, std=5.7921, median=0.0000, min=0.0000, max=23.1600
scenario 2: n=400, mean=7.7933, std=16.9026, median=0.0000, min=0.0000, max=90.4100

Deliverables

- simulate_auction.py
- data_exp4_auction.csv
- regression_results.csv
- overbid_summary.csv
- run_summary.txt
- analysis.do
- analysis.prg
- auction_flowchart.svg
- regression_results.png
- scenario_bar.png
- scenario_box.png
- report.txt
- report.pdf